Solar Irradiance Prediction: An Artificial Neural Network Approach versus Modified Pro-Energy

Deep Learning Optimization for Energy-Harvesting Wireless Sensor Networks

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Research Overview

The Core Question

Does a data-driven Artificial Neural Network (ANN) outperform the state-of-the-art rule-based Modified Pro-Energy method for solar irradiance prediction in WSNs?

- **Methodology**: Comparative analysis of ANN vs. classical baselines (EWMA, WCMA, MPE).
- **Dataset**: Full year NREL measurements (2019), 266,267 raw samples.
- **Key Result**: ANN achieves $R^2 = 0.9867$ and $RMSE = 20.027\ W/m^2$.
- **Impact**: Nearly 4-fold performance improvement over Modified Pro-Energy on high-variance days.

Nomenclature

C	Harvested energy (current day) $[W/m^2]$
N	Total number of time-slots in a day
D	Stored past day energy profile
MAE	Mean absolute error [%]
E	Matrix to store profiles ($D \times N$)
P	Number of energy profiles to identify
WP	Weight profile within combined P
H	Actual energy in previous slots
MD	Mean value of past day energy samples
$E(d,n)$	Predicted energy (day d, slot n)
E_L	Battery energy level $[J]$
$\rho(d,n)$	Expected error [%]
$H(d-1,n)$	Past day energy $[W/m^2]$
$I_{\text{solar-cell}}$	Node fetches irradiance value
α	Weighting factor for prediction
$E(d-1,n)$	Expected past day energy

Motivation: Sustainable EH-WSNs

- **WSN Context**: Remote deployments (IoT, Environmental tracking) where battery replacement is infeasible.
- **Solar Harvesting**: Photovoltaic (PV) panels + Rechargeable batteries.
- **The Predictive Gap**: Without accurate forecasts, Power Management Units (PMU) cannot optimize duty cycles or transmission power.
- **Resource Constraint**: Algorithms must run on low-power microcontrollers (limited memory/CPU).

The Challenge

Solar energy is stochastic and highly dependent on time-of-day, season, and atmospheric conditions. Efficient prediction is the key to node longevity.

Physical Motivation: The PV Cell Model

- **Non-Linear Dynamics**: PV harvesting depends on irradiance, temperature, and electrical load.
- **The Critical Driver**: Irradiance (W/m^2) is the fundamental physical quantity driving power output.
- **Approach**: Treat irradiance as the primary prediction target to capture the stochastic environment.

The Prediction Task

Estimate the expected energy $\hat{E}(d, n)$ for the next time slot using historical measurements, minimizing cumulative error:

$$\bar{\varepsilon} = \frac{1}{D \cdot N} \sum_{d=1}^{D} \sum_{n=1}^{N} \left| \hat{E}(d, n) - H(d, n) \right| \tag{1}$$

Goal: Balance accuracy (prevent outages) with efficiency (avoid waste).

Evolution of Prediction Methods

1. **EWMA (Exponential Weighted Moving Average)**:

$$E(d,n) = \alpha E(d-1,n) + (1-\alpha)H(d-1,n) \tag{2}$$

✗ Simple, but carries memory of only one past day; no weather model.

2. **WCMA (Weather-Conditioned Moving Average)**:

$$E(d,n) = \alpha H + (1-\alpha)M_D(d,n) \cdot GAP_k \tag{3}$$

✓ Uses a dimensionless "GAP" coefficient to track current weather trends.

3. **Pro-Energy (Profile Energy)**: Retrieves canonical profiles (Sunny, Cloudy, Mixed) via MAE matching. ✗ "Historical Lock": Fails if today's weather has no analogue in the library.

The Paradigm Shift: Mapping over Lookup

Rule-Based (Lookup)

- Memory-intensive profile libraries.
- Static matching templates.
- Struggles with multi-scale patterns.
- High error on "Novel" days.

Deep Learning (ANN)

- Learns the **Mapping Function**: $f(\text{time, lag, sensor}) \rightarrow \text{GHI}$.
- Generalizes to unseen weather combinations.
- Lightweight parameter storage (no profile DB).
- **Reactive**: Real-time adaptation via lag features.

Dataset Insights: Solar Irradiance Distribution

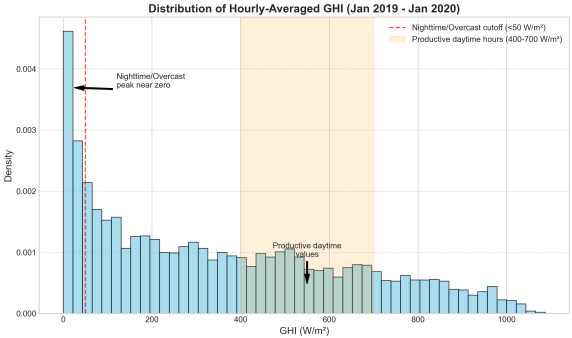


figure : Bimodal structure: night vs day.

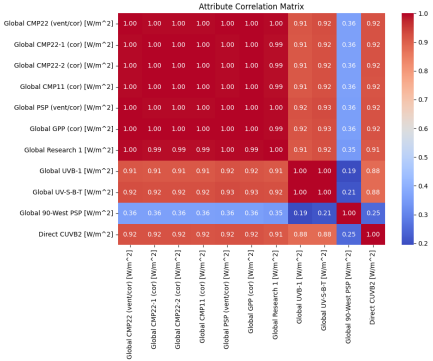
1	datetime	Global CMP22	Global UVB-1 [W/m^2]	Global 90-West PSP [W/m^2]	Direct CUVB2 [W/m^2]
2	2019-01-26 07:00:00	24.05	0.00	15.74	0.00
3	2019-01-26 08:00:00	125.37	0.05	76.24	0.02
4	2019-01-26 09:00:00	381.46	0.18	199.26	0.20
5	2019-01-26 10:00:00	430.22	0.31	217.49	0.15
6	2019-01-26 11:00:00	541.48	0.45	211.03	0.36
7	2019-01-26 12:00:00	582.79	0.49	261.51	0.46
8	2019-01-26 13:00:00	534.78	0.40	454.48	0.39
9	2019-01-26 14:00:00	422.07	0.25	595.61	0.26
10	2019-01-26 15:00:00	281.61	0.10	609.38	0.12
11	2019-01-26 16:00:00	80.62	0.02	204.49	0.00
12	2019-01-26 17:00:00	11.39	0.00	9.75	0.00
13	2019-01-27 07:00:00	67.72	0.01	40.77	0.00
14	2019-01-27 08:00:00	177.22	0.05	82.93	0.04
15	2019-01-27 09:00:00	289.57	0.18	123.35	0.11
16	2019-01-27 10:00:00	244.89	0.24	115.36	0.01

figure : Training/Testing Split Glimpse.

- **Bimodal Structure**: High concentration near zero (night) and 400–700 W/m^2 (day).
- **High Variance**: Peak irradiance can exceed 1,000 W/m^2, requiring robust scaling.

Data Preprocessing Pipeline

- **Temporal Aggregation**: 1-minute NREL readings averaged into hourly windows.
- **Feature Pruning**: Reduced 11 redundant sensor channels to 4 distinct physical inputs based on **Pearson Correlation** (> 0.98).



Physical Targets:

1. Global CMP22 (*GHI*)
2. Global UVB-1
3. Global 90-West PSP
4. Direct CUVB2

Physics-Informed Feature Engineering

Cyclic Temporal Encoding

Integer time (0–23) creates artificial discontinuities. We map time onto a unit circle:

$$\sin_{\text{hour}} = \sin\left(\frac{2\pi \cdot \text{hour}}{24}\right), \quad \cos_{\text{hour}} = \cos\left(\frac{2\pi \cdot \text{hour}}{24}\right) \tag{4}$$

Ensures adjacent hours (23 and 0) are geometrically close.

Autoregressive Lag Feature

$$\text{GHI_lag}_1(t) = \text{GHI}(t-1) \tag{5}$$

Captures atmosphere momentum; provides the model a "running start" for next-hour extrapolation.

Lightweight ANN Architecture

- **Design Rationale**: Minimize overfitting for small datasets (4,419 samples) while ensuring edge compatibility.
- **Structure**:
  - **Input**: 7 engineered features.
  - **Hidden 1**: 64 ReLU neurons ($\mathbf{h}^{(1)} = \text{ReLU}(\mathbf{W}^{(1)}\mathbf{x} + \mathbf{b}^{(1)})$).
  - **Hidden 2**: 32 ReLU neurons.
  - **Output**: 1 Linear neuron (GHI prediction).
- **Optimization**: Adam Optimizer ($\beta_1 = 0.9, \beta_2 = 0.999$) with MSE loss.

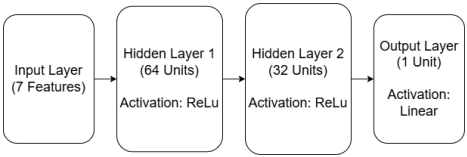


figure : Compact Feed-Forward Blueprint

The State-of-the-Art Baseline: Modified Pro-Energy

- **Mechanism**: Blends profiles weighted by similarity (MAE):

$$W_j = 1 - \frac{MAE(E_{d_j}, C)}{\sum_{j=1}^{P} MAE(E_{d_j}, C)} \tag{6}$$

- **Error-Correction (ρ)**:

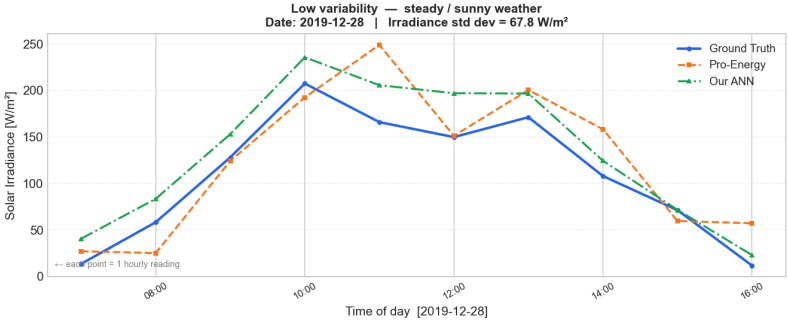
$$\rho(d, n) = \beta\rho(d, n-1) + (1-\beta)\left[\hat{E}(d, n-1) - E_{\text{actual}}(d, n-1)\right] \tag{7}$$

- **Inference Complexity**: $O(D \times N)$ per day.
- **Overhead**: Higher RAM burden due to $D \times N$ matrix maintenance.

Quantitative Performance Matrix

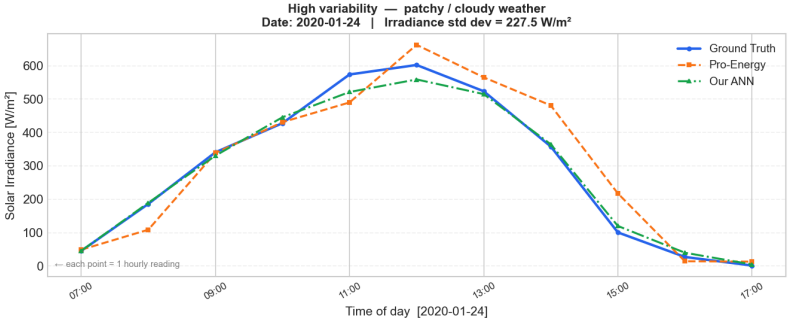
Metric	ANN (Proposed)	MPE	EWMA	WCMA	Std Pro-E
RMSE (W/m^2)	**20.027**	73.721	124.273	96.087	152.262
MAE (W/m^2)	**19.910**	51.552	61.050	51.402	85.670
R^2 Score	**0.9867**	0.8203	0.7977	0.8793	0.6959

Performance Analysis: The Clear Sky Scenario



- On stable days (e.g., Dec 28), both models track ground truth effectively.
- **Profile-Matching** is correct by construction on clear days.
- ANN provides slightly better alignment near the peak.

Performance Analysis: The High-Variance Scenario



- **The Asymmetry**: Modified Pro-Energy locks onto the wrong profile early and fails to adjust to cloud transients.
- **ANN Reaction**: Instantaneous processing of the current feature vector allows it to follow transitions without systematic offsets.

Why the ANN Wins: The "Historical Lock" Problem

Modified Pro-Energy

Makes a "bet" at the start of the day. If weather shifts mid-day, the correction factor (ρ) is insufficient to compensate for an fundamentally wrong profile choice.

Artificial Neural Network

Re-evaluates from the current lag feature every hour. It does not "expect" a pattern; it simply computes the next most likely state based on immediate evidence.

Insights from short-term vs long-term predictions

- **Short-term (minute-level) prediction is easier** because solar irradiance changes gradually, leading to high similarity between consecutive values.
- **Simple methods perform well at fine granularity** (e.g., persistence, EWMA, WCMA) since they rely on recent values and benefit from strong temporal autocorrelation.
- **Hourly prediction is significantly harder** due to larger changes caused by clouds, weather shifts, and day–night transitions.
- **WSN applications require long-term forecasts**, as nodes must plan energy usage, routing, and duty cycles hours in advance.

Insights from short-term vs long-term predictions

- **Data volatility increases sharply at hourly level**, with variance ($\sim 43\times$), mean absolute change ($\sim 11\times$), and standard deviation ($\sim 6.5\times$) all much higher, along with larger sudden jumps.
- **Traditional methods struggle with this variability**, whereas ANN models can capture complex, non-linear patterns and perform more reliably on unstable hourly data.

Table: Statistical Metrics of Signal Volatility ($N = 1000$)

Metric	**Minute-Level**	**Hourly-Level**	**Ratio (H/M)**
Variance of Change	535.07	22848.30	42.70
Volatility (Std Dev)	23.13	151.16	6.53
Mean Abs Change	11.33	121.99	10.77
Max Jump	175.93	720.66	4.10

Conclusion & Future Outlook

- **Validated**: Data-driven mapping (ANN) is fundamentally superior to template matching (Pro-Energy) for non-stationary irradiance.
- **Efficient**: Compact architecture allows practical deployment on EH-WSN nodes.
- **Next Steps**:
  1. **Cross-Geographic Training**: Evaluating tropical vs arid performance.
  2. **Protocol Integration**: Linking prediction directly to MAC scheduling and Cluster Head selection.
  3. **Multi-Step Forecasting**: Extending the horizon beyond next-hour.

Thank You! Questions?